

# Lectures

Stat 154/254: Statistical Machine Learning

Lectures marked with ★ are still in progress.



## Introduction

- Class introduction
- Key concepts

## Unit 1: Regression in a machine learning setting

- Linear regression in matrix notation (review)
- Linear regression in an ML setting
- Basis expansions and splines for linear regression
- L1 and L2 penalization

## Unit 2: Classification

- Empirical risk minimization for classification
  - Perceptron algorithm (**extra, not covered in class**)
- Trading off false positives and negatives

## Unit 3: Risk and complexity

- Regret decomposition
- Asymptotics and bias–variance tradeoff in linear regression
- Generalization bounds
  - Generalization error and uniform laws for smooth function classes
  - VC dimension and zero–one loss
- Cross validation
  - ★ Held-out data and cross-validation
  - ★ Simulated bias–variance tradeoff example
  - ★ Cross-validation cautionary note

## Unit 4: Trees and weak learners

- ★ Classification and regression trees
- ★ Bagging weak learners
- ★ Boosting weak learners

## Unit 5: Neural networks and optimization

- [★ Optimization notes](#)

### Prerequisites

Mathematical statistics, probability, linear algebra, multivariate calculus, and Python are all prerequisites.

For assistance with mathematics prerequisites, the [Student Learning Center](#) provides additional assistance.

A summary of the required linear algebra skills can be found [here](#).